

BREATHEBOX · CALCULATION NOTE

BreatheBox sizing calculations

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
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REVISION HISTORY

Version	Date	Author	Change
0.7	2026-10-02	Amish Chadha	Approved follow-ups carried out: wall foot at 400 mm (sill friction 0.32, margin 1.9; strut joints rerun), sash jammer added as BOM line 14 (R12), mass 11.8 kg installed, cost USD 294 (USD 39 over the value-engineering target); fan position decided
0.6	2026-10-02	Amish Chadha	R12 text: sash held by a sash jammer, decided by Amish on 2026-10-02; no figures changed
0.5	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.4	2026-09-30	Amish Chadha	Constructable design (BBX-DDR-003); mass from the components as made, sill stability and bracket joints, filter seat and grille losses, cost \$283 so R15 is not met
0.3	2026-09-26	Amish Chadha	Budget approved by Amish; budget \$255 covers the priced BOM, so R15 is met (BBX-DDR-002 v0.2)
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); budget \$250, quiet night

mode at 32 m³/h and firmware rules added

0.1

2026-09-25

Amish Chadha

First TRL 3 sizing note (core, pressure drop, fans, power, noise, heat recovery, condensate, frost, CO₂, geometry, mass, cost) against BBX-REQ-001 v0.3

BreatheBox sizing calculations

On paper, BreatheBox meets ten of its fifteen requirements. **One is not met and one is over its value-engineering target.** R6 (noise): the estimate is about 39 dB(A) at 1 m at 50 m³/h against 30 dB(A), and 30 dB(A) is reached only at about 32 m³/h. A quiet night mode at 32 m³/h (BBX-DDR-002) brings the estimate to about 30 dB(A) with clean filters and 32 dB(A) with loaded filters, but R6 is written for 50 m³/h. R15 (cost): Value-engineering target: USD 255. Estimated cost of the constructable design: USD 294 (USD 39 over the target), including the sash jammer decided on 2026-10-02; the savings worth trying are in the design decisions register (BBX-DEC-001). Three requirements (R9 install time, R13 filter change time and R14 firmware behavior) cannot be verified at TRL 3.

The calculations changed four TRL 2 figures. The TRL 2 fans (about 60 m³/h free air) could not reach 70 m³/h against any pressure, so the fan specification rises to about 100 m³/h free air and 400 Pa shut-off. Frost starts at about -2 °C outdoors, not -5 °C, because the plate at the cold corner is colder than the leaving exhaust air. The worst condensate flow is about 0.21 L/h, not 0.14 L/h. Fan and control power at 50 m³/h is about 5.6 W, not 9.5 W, because the system pressure is lower than the 80 Pa assumed.

Every number here is printed by `docs/04-calcs/sizing.py` (run from the repo root: `python docs/04-calcs/sizing.py`), which also writes `docs/04-calcs/results.csv`. The script reads the geometry and part volumes from `cad/src/model.py`, the costs from `bom/bom.csv` and the budget from `project.yaml`. All values are first-principles estimates; nothing is measured.

■ 1. Assumptions

Table 1. Inputs. All are assumptions for a paper design.

Input	Value	Basis
Design case	30 m ³ bedroom, two sleeping adults, 20 °C and 50 % RH indoors, 0 °C outdoors	BBX-REQ-001
Air	1.2 kg/m ³ , 1,005 J/(kg·K), 1.5×10^{-5} m ² /s, 0.0257 W/(m·K)	Near 20 °C
Airflow per stream	30 to 70 m ³ /h, 50 m ³ /h nominal; 32 m ³ /h in night mode	BBX-REQ-001 R1; BBX-DDR-001 item A5; BBX-DDR-002
Core	180 x 180 x 300 mm, 2.5 mm plate pitch, 0.15 mm polymer plates (0.19 W/(m·K)); 85 % of the plate area acts in counterflow	Typical small polymer plate core; no datasheet yet
Core heat transfer	Fully developed laminar flow between plates, Nu = 7.54; headers add 2.0 dynamic heads	Textbook laminar values
Supply filter	ISO 16890 ePM1 50 %, 25 mm pleated, 150 x 170 mm face; 50 Pa at 1.0 m/s clean, linear with velocity; replaced at twice the clean pressure drop ("loaded")	Typical for this filter class; to be replaced by the chosen filter's data
Exhaust filter	Coarse pad, 220 x 190 mm; 15 Pa at 1.0 m/s clean; loaded at twice	Typical
Hood and mesh	Hood turn K = 1.5, 1 mm mesh (open area about 59 %) K = 2.0 at low Reynolds number, on the 124 x 174 mm mouth	Conservative loss coefficients
Other losses	Port and collar K = 1.0; plenum turn K = 1.5 on the half core face; fan inlet K = 1.0 on a 68 mm inlet; grilles K = 1.5 on 50 % free area	Conservative loss coefficients
Fans	Full-speed curve $\Delta p = 400 \text{ Pa} \times (1 - (Q/100 \text{ m}^3/\text{h})^2)$; speed scales by the fan laws; total efficiency 25 %, plus 0.3 W electronics per fan	Class values for 120 x 120 x 32 mm, 24 V centrifugal blowers; no datasheet yet
Fan noise	52 dB(A) at 1 m for a bare fan at full speed; 50 log ₁₀ of the speed ratio; 5 dB taken off for the lined plenum and grilles	Class value and fan law; the largest uncertainty in this note
Controls	0.5 W at 3.3 V through an 85 % buck	ESP32-C3 class module with an SCD41 class sensor
CO2	0.014 m ³ /h per sleeping adult, 420 ppm outdoors, no infiltration, 8 h of sleep	Common planning values
Season	3,500 K·d over 212 days	Cold-temperate climate
Depressurization	5 Pa screening limit; 800 x 10 mm door undercut, discharge coefficient 0.6	Screening assumption; to be checked against local code
Densities	PVC foam board 550 kg/m ³ ; closed-cell foam 30 kg/m ³ ; 10 x 10 mm strip 550 kg/m ³ ; panel 190 kg/m ³ effective; rigid PVC sheet 1,400 kg/m ³ ; PETG 1,270 kg/m ³ ; aluminium 2,700 kg/m ³ ; modelled bolts 7,900 kg/m ³ ; applied to each component's modelled volume	Typical materials
Sill friction	About 0.6 for anti-slip rubber tape on a painted or varnished sill	Assumption; to be checked with a push test

■ 2. Core effectiveness (R2)

The core has 72 channels (36 per stream) with a 2.35 mm gap and a 4.7 mm hydraulic diameter, and 71 plates with 3.83 m² of area, 3.26 m² of it effective. The laminar film coefficient is 41.2 W/(m²·K) on each side, so U is 20.3 W/(m²·K) and UA is 66.1 W/K. Because the flow in the channels stays laminar (Reynolds number 171 to 400), UA does not change with airflow.

Table 2. Core performance, balanced flow.

Airflow	Heat capacity rate	NTU	Effectiveness	Channel velocity	Core pressure drop
30 m ³ /h	10.05 W/K	6.58	86.8 %	0.55 m/s	6.8 Pa
50 m ³ /h	16.75 W/K	3.95	79.8 %	0.91 m/s	11.7 Pa
70 m ³ /h	23.45 W/K	2.82	73.8 %	1.28 m/s	16.9 Pa

R2 is met on paper: 79.8 % at 50 m³/h against 75 %. At the 70 m³/h boost the effectiveness falls to 73.8 %, which is outside R2's 50 m³/h case.

■ 3. System pressure drop (R1)

The supply stream loses more than the exhaust stream because it carries the ePM1 filter. The filter and the fan inlet dominate; the hoods, mesh and ports add less than 3 Pa together.

Table 3. Pressure drop per stream, clean and loaded filters.

Airflow	Supply, clean	Supply, loaded	Exhaust, clean	Exhaust, loaded
30 m ³ /h	27.9 Pa	44.3 Pa	14.4 Pa	18.1 Pa
50 m ³ /h	52.3 Pa	79.6 Pa	28.7 Pa	34.8 Pa
70 m ³ /h	81.4 Pa	119.5 Pa	46.8 Pa	55.4 Pa

At 50 m³/h the clean supply stream splits into filter 27.2 Pa, core 11.7 Pa, fan inlet 8.8 Pa, room grille 2.9 Pa, hood and mesh 0.9 Pa, plenum turn 0.7 Pa and port 0.2 Pa. The TRL 2 estimate of 80 Pa matches only the loaded supply filter case. The constructable design (BBX-DDR-003) adds about 0.5 Pa on the supply side, because the top grille opening is now 70 x 220 mm over the supply fan's outlet chamber, and about 1.1 Pa on the exhaust side, because the exhaust pad works through a 200 x 170 mm lining lip. The supply filter's lip lies on its frame, not its media, so its drop is unchanged.

■ 4. Fans, power and noise (R1, R3, R6, R10)

Operating points. With the assumed full-speed curve, the fans reach 86 m³/h per stream with clean filters and **80 m³/h with loaded filters**, so the 30 to 70 m³/h range of R1 is covered with a margin. The TRL 2 fan (about 60 m³/h free air) could not reach 70 m³/h even with no system resistance, which is why BOM items 3 and 4 now call for about 100 m³/h free air and 400 Pa shut-off. The supply and exhaust fans run at different speeds for the same flow (0.62 and 0.57 of full speed at 50 m³/h), so balance within 10 % needs the tach feedback and a one-time calibration against the system curves. **R1 is met on paper.**

Table 4. Operating points. Power is fans plus controls.

Case	Supply speed	Exhaust speed	Power	Room noise at 1 m
30 m³/h, clean	0.40	0.35	2.6 W	29.0 dB(A)
30 m³/h, loaded	0.45	0.37	3.3 W	30.9 dB(A)
50 m³/h, clean	0.62	0.57	5.7 W	38.7 dB(A)
50 m³/h, loaded	0.67	0.58	7.5 W	40.0 dB(A)
70 m³/h, clean	0.83	0.78	11.2 W	45.4 dB(A)
70 m³/h, loaded	0.89	0.79	14.8 W	46.4 dB(A)

Power (R3). 5.7 W at 50 m³/h with clean filters and 7.5 W with loaded filters, against 12 W. **R3 is met on paper.** Over the heating season the fans and controls use about 29 kWh.

Noise (R6). At 50 m³/h the estimate is about **39 dB(A) at 1 m, so R6 is not met.** With clean filters the unit reaches 30 dB(A) only at about 32 m³/h. The noise figure rests on an assumed catalog value and a fan law, so it may be several decibels out in either direction, but even a 5 dB error leaves R6 unmet at 50 m³/h.

Night mode (BBX-DDR-002). Amish accepted the recommendation of a quiet night mode, with larger, slower fans or a lined silencer to be checked against a real fan datasheet before any change to R6. The firmware rule is: in night mode both streams run at 32 m³/h (balanced by tach), the CO2 boost is suppressed, and frost mode and the speed cap still apply. At 32 m³/h the estimate is **30.2 dB(A) with clean filters and 32.0 dB(A) with loaded filters**, for 2.8 W and 3.6 W. With loaded filters, 30 dB(A) would need about 28 m³/h, below the 30 m³/h floor of R1, so the night mode stays at 32 m³/h and relies on filter changes. The datasheet check is on hold with TRL 4.

Adapter and fuse (R10). The 70 m³/h loaded case needs 14.8 W (0.62 A at 24 V). The worst case, both fans at full speed at their peak air power (58 m³/h each on the assumed curve), is about 35.4 W (1.48 A), just under the 36 W adapter rating. The firmware caps fan speed at what 80 m³/h needs (a firmware rule accepted in BBX-DDR-002), and the 24 V input carries a 2 A time-delay fuse. **R10 is met by design** (24 V SELV only).

■ 5. Heat recovery

Table 5. Heat at 0 °C outdoors and 20 °C indoors.

Airflow	Heat carried by exhaust	Recovered	Lost	Supply air
50 m³/h	335 W	267 W	68 W	16.0 °C (16.1 °C with fan heat)
70 m³/h	469 W	346 W	123 W	14.8 °C (15.0 °C with fan heat)

Over a season of 3,500 K·d at 50 m³/h the core keeps about 1,123 kWh of heat in the room for about 29 kWh of fan and control energy, a ratio of about 39 to 1. The 335 W is also the heat an open window would lose at the same airflow.

■ 6. Condensate and frost (R7)

Condensate. Room air at 20 °C and 50 % RH holds 7.3 g/kg. The upper bound assumes the exhaust cools to its dry outlet temperature and leaves saturated. At 0 °C outdoors this gives 132 g/h at 50 m³/h (exhaust leaves at 4.0 °C) and 148 g/h at 70 m³/h. The worst case above frost onset is **214 g/h (0.21 L/h) at 70 m³/h and -3.0 °C**, above the 0.14 L/h estimated at TRL 2. That is 0.06 mL/s, which an 8 mm bore tube falling outward drains easily; the risk is the tube freezing outdoors, not its size.

Frost onset. Frost forms where the plate is coldest: the corner where cold supply air enters and exhaust air leaves. With equal film coefficients on both sides, that plate sits midway between the two air temperatures. Frost starts at about **-2.2 °C outdoors at 50 m³/h** and -3.0 °C at 70 m³/h. The TRL 2 figure of -5 °C used the exhaust air temperature alone and was too optimistic.

Frost mode. Slowing the supply fan (BBX-DDR-001 item A4) keeps the cold corner at 0 °C at -10 °C outdoors when the supply flow is 33 % of the exhaust flow: 17 m³/h against 50 m³/h. The room still receives 50 m³/h of fresh air, 17 m³/h through the core and 33 m³/h drawn from the rest of the home. Through an 800 x 10 mm door undercut that imbalance depressurizes the room by about 2.2 Pa (4.4 Pa at 70 m³/h), under the 5 Pa screening limit. With the door sealed, the depressurization could be much larger. **R7 is met on paper**, with the combustion-appliance restriction in BBX-PRC-001.

A preheater that kept the flows balanced at -10 °C would need about 130 W, more than three times the 36 W adapter. This supports the slowed-fan strategy for a 24 V unit.

Safety: Frost mode extracts more air than it supplies. Do not use BreatheBox in a room with an open-flued or unflued fuel-burning appliance, and do not install it in a room whose door seals airtight.

■ 7. CO₂ (R4)

Table 6. CO₂ with two sleepers in a 30 m³ room, no infiltration.

Airflow	Night steady state	24 h mean (8 h occupied)	Time constant
30 m ³ /h	1,353 ppm	731 ppm	60 min
50 m ³ /h	980 ppm	607 ppm	36 min
70 m ³ /h	820 ppm	553 ppm	26 min
32 m ³ /h, night mode all day (bounding case)	1,295 ppm	712 ppm	56 min

R4 is met on paper: the 24 h mean is 607 ppm at 50 m³/h against 1,000 ppm, and even the night steady state (980 ppm) stays under 1,000 ppm. At 30 m³/h the 24 h mean still meets R4, although the night level reaches about 1,350 ppm. This matters for R6: the night mode at 32 m³/h (BBX-DDR-002) meets R4 as written, with a 24 h mean of 712 ppm even if it ran all day, while the room air rises toward about 1,300 ppm for part of the night.

■ 8. Geometry, structure and mass (R8, R9, R11, R12)

- **R11, met on paper.** The hood mouths are 130 mm wide with outer edges at ±340 mm, leaving 420 mm between the supply intake and the exhaust outlet. Both face down, with 1 mm mesh.

- **R8, met on paper for sash windows.** The hoods need at least 686 mm of clear width, just inside the 700 mm lower bound; the panel trims to 700 to 1,000 mm. Casement and tilt-and-turn windows remain outside this version (BBX-DDR-001 item A2).
- **R12, met by design.** The largest opening through the installed unit is a 2.35 mm core channel; the mesh is 1 mm. The raised sash is held down on the panel by a bought no-drill sash jammer wedged in the inner track between the top of the raised sash and the window head (decided 2026-10-02, BBX-DEC-001; bill of materials line 14, about \$10, 0.30 kg estimated).
- **Housing.** 510 x 560 x 250 mm, with 110 mm on the sill and 400 mm projecting into the room. The model now holds every component as it is made (BBX-DDR-003); its 90 constructability checks pass and no two of its 41 components (the sash jammer included) overlap.
- **Stability on the sill (BBX-DDR-003).** The housing, its two rails, the struts and the wall foot are one rigid body, because each strut end has two bolts. The parts it carries weigh 8.6 kg with their centre of mass 162 mm on the room side of the wall face. The body rests on the inner edge of the sill and pushes on the wall at the foot, whose centre line is 400 mm above the floor (500 mm below the sill edge), with about 27 N; the collars only slide into the housing, so friction at the sill is all that stops it sliding into the room. The friction needed is 0.32, against about 0.6 for anti-slip tape: a margin of about 1.9. With the foot at 500 mm it would be 0.40 (margin 1.5), and with the concept's foot at 650 mm, 0.65. Amish decided on 2026-10-02 to build the first prototype with the foot at 400 mm and to run a 50 N push test at the trial window (BBX-DEC-001); the foot goes back to 500 mm only where a radiator or skirting blocks the lower foot and the push test passes on that sill. The struts are now 595 mm between end holes (cut 621 mm), rising at 55°.
- **Bracket joints.** Each strut's top joint carries about 6.6 N·m: 18 MPa of bending in the 20 x 1.5 mm tube (6063 yields at about 110 MPa) and about 329 N of shear on each of its two M6 bolts. Both are small. The lower foot pushes less on the wall, but the longer struts carry that push over a longer arm, so the joint figures are almost unchanged.

Table 7. Mass from the modelled components (BBX-DDR-003).

Item	Mass
1 Housing: boards, battens, lining, core frames, dividers, bulkhead, filter seats, lid, grilles, latches	4.79 kg
2 Core (plates 0.78 kg, frame 0.30 kg)	1.08 kg
3, 4 Fans	0.60 kg
5, 6 Filters	0.15 kg
7 Controller and status board	0.08 kg
8 Tray, pads and drain	0.24 kg
9 Insert panel and seals	0.70 kg
10 Collars, hoods and back plates	2.08 kg
11 Sill bracket: rails, cleats, struts, foot and pad	1.27 kg
12 Adapter	0.25 kg
13 Modelled bolts and cable gland	0.19 kg
14 Sash jammer (bought; estimate)	0.30 kg
Sundries (wiring, gasket, glue, small screws)	0.30 kg
Total	12.0 kg (11.8 kg installed, without the adapter)

The mass part of R9 is met on paper, with a small margin: 11.8 kg against 12 kg (11.5 kg without the sash jammer). It was 11.4 kg before the 2026-10-02 follow-ups (longer struts and the sash jammer) and 10.5 kg before the design was made constructable. The 30 min install time cannot be verified until someone fits the unit, so **R9 is not verifiable at TRL 3**.

■ 9. Cost (R15)

Value-engineering target: USD 255. Estimated cost of the constructable design: USD 294 (USD 39 over the target). The priced BOM totals **\$294.00** over 14 lines; the target is `budget_usd`, set on 2026-09-26 (BBX-DDR-002; it was \$250, and \$220 before 2026-09-25), and is unchanged. **R15 is over the value-engineering target by \$39**. The concept's BOM was \$255; the rise comes from parts the concept needed but did not list, added when the design was made constructable (BBX-DDR-003): partitions, filter seats, grilles and latches (+\$18), hood back plates and collar flanges (+\$2), the bracket's rails, cleats and foot (+\$4), and bolts, gland and anti-slip tape (+\$4). The 2026-10-02 decisions added the sash jammer (line 14, +\$10) and longer struts for the 400 mm foot (line 11, +\$1). The savings worth trying are in the design decisions register (BBX-DEC-001). Earlier, the increase from \$243 to \$255 came from the larger fans (+\$8), the collars through the panel (+\$2) and the input fuse and holder (+\$2).

■ 10. Results

Table 8. Requirement status (from `docs/04-calcs/results.csv`).

ID	Value	Target	Status
R1	30 to 70 m³/h per stream; full-speed limit 80 m³/h with loaded filters; balanced by tach	30 to 70 m³/h, within 10 %	Met on paper
R2	79.8 % at 50 m³/h (73.8 % at 70 m³/h)	75 % or more at 50 m³/h	Met on paper
R3	5.7 W clean, 7.5 W loaded	12 W or less at 50 m³/h	Met on paper
R4	24 h mean 607 ppm (712 ppm in night mode); night 980 ppm (1,295 ppm in night mode)	1,000 ppm or less, 24 h mean	Met on paper
R5	ePM1 50 % supply, coarse exhaust	ePM1 50 % or better	Met by specification
R6	39 dB(A) at 50 m³/h; night mode at 32 m³/h 30 dB(A) clean, 32 dB(A) loaded	30 dB(A) or less at 50 m³/h	Not met
R7	Worst condensate 0.21 L/h drains; frost onset -2.2 °C; supply at 33 % of exhaust at -10 °C	Drain all condensate; no blockage to -10 °C	Met on paper
R8	Hoods need 686 mm clear width; panel 700 to 1,000 mm	Sash 700 to 1,000 mm; slider adapter	Met on paper (sash)
R9	11.8 kg (sash jammer included); no drilling; install time not calculable	12 kg or less; 30 min or less	Not verifiable at TRL 3 (mass met on paper)
R10	36 W SELV adapter; 14.8 W at 70 m³/h loaded; 35.4 W worst case; 2 A fuse	24 V SELV only	Met by design
R11	420 mm between mouths; 1 mm mesh; mouths face down	400 mm or more; mesh 1.5 mm or finer	Met on paper
R12	Largest opening 2.35 mm; raised sash held down on the panel by a bought no-drill sash jammer (BOM line 14)	Sash lockable onto the panel; no opening over 100 mm	Met by design, unverified
R13	Lift-off lid; filters and core slide out	Tool-free filter change in 2 min or less	Not verifiable at TRL 3
R14	No firmware yet; checked at firmware review	Local data only	Not verifiable at TRL 3
R15	\$294.00	\$255 value-engineering target	Over the value-engineering target by \$39

Summary: met 10, not met 1 (R6), over the value-engineering target 1 (R15), not verifiable at TRL 3 3 (R9, R13, R14).

■ 11. Limits of this note

- The fan curve, fan noise and filter resistance are class values, not datasheet values. R1, R3, R6 and R10 should be rechecked when parts are chosen; the night mode noise figure carries the same uncertainty.
- Laminar, fully developed flow and equal film coefficients are idealizations; real cores with headers usually recover a few points less than this model predicts.
- Leakage between streams in the core, air leakage around the insert panel and the latent heat released by condensation are not modeled. The last makes the frost onset slightly conservative.

- In the chosen layout both fans sit in the room-end plenum, so the exhaust side of the core runs at a higher pressure than the supply side. Any core leakage would carry stale air into the supply. Amish decided on 2026-10-02 to keep both fans at the room end for the prototype; a cross-leakage check (CO2 tracer in the exhaust stream, measured in the supply) is part of the TRL 4 checks, and the exhaust fan moves to the outdoor end only if the transfer exceeds about 3 % (BBX-DEC-001).